

T. VINOTH VELIAPPAN

Senior Cloud & DevOps Engineer

+91 98403 68580 | hivinoth8@gmail.com | Chennai, Tamil Nadu | [linkedin.com/in/vinoth](https://www.linkedin.com/in/vinoth)
AWS Certified Solutions Architect – Professional (Jan 2026)

PROFESSIONAL SUMMARY

Results-driven Senior Cloud & DevOps Engineer with 10+ years of IT experience, including 6+ years architecting and operating production AWS environments. Deep expertise in ECS/EKS container orchestration, multi-tier network architecture, CI/CD GitOps automation, AI/ML Ops (vLLM, Ollama), and full-stack observability. AWS Certified Solutions Architect – Professional with a proven track record of reducing release cycles by 60%, cutting infrastructure costs, and improving system reliability at scale.

CORE COMPETENCIES

Cloud & IaC	AWS (EC2, ECS, EKS, S3, RDS, ALB, CloudFront, WAF, Route 53, CloudFormation, Terraform)
Containers & Orchestration	Docker, Amazon ECS (Fargate/EC2), Amazon EKS, Kubernetes, Cloud Map Service Discovery
Network Architecture	WAF → CDN (CloudFront) → ALB → Nginx (ECS/EKS) → Cloud Map → Microservices; VPC, Subnets, NACL, SG, Transit Gateway
CI/CD & GitOps	GitHub Actions, GitLab CI, Jenkins, Argo CD, AWS CodeDeploy, Bamboo
AI/ML Ops	vLLM (Hugging Face, PagedAttention), Ollama (GGUF – Llama 3, Mistral)
Observability	Prometheus, Grafana, CloudWatch, EFK Stack, OpenSearch, Filebeat, Kibana, OpenObserve, Airbrake
Security & Governance	AWS WAF, GuardDuty, Security Hub, IAM, AWS Control Tower, Keycloak, CIS Benchmarks
Databases	MySQL, PostgreSQL, MongoDB, RDS, Amazon OpenSearch
Data & Streaming	AWS Glue, DMS, Athena, Lake Formation, Kafka, Weaviate
Web Servers & LB	Nginx, Apache, Tomcat, HAProxy, AWS ALB/ELB, Nginx Ingress
Other Tools	GrowthBook, Airbrake, AWS Transfer Family (SFTP), ECR, SonarQube

KEY ACHIEVEMENTS

- Architected end-to-end traffic flow (WAF → CloudFront/S3 → ALB → Nginx ECS → Cloud Map → microservices), improving security posture and enabling zero-downtime deployments.
- Designed AWS infrastructure for 80+ microservices on EKS/ECS, reducing API latency by 35% via cluster-level optimizations and right-sizing.
- Deployed Hugging Face models on vLLM with PagedAttention, achieving 4× throughput (1,200 → 4,800 req/sec) and 40% GPU cost reduction on H100 instances.
- Automated EKS/ECS deployments with Argo CD & GitHub Actions, cutting release cycles from 2 hours → 15 minutes and eliminating manual deployment errors.
- Migrated 12 legacy applications to AWS using CloudEndure — 90% reduction in downtime and \$5K/year colocation cost savings.
- Built Prometheus/Grafana observability stack, reducing MTTR by 60% via real-time anomaly detection.
- Centralized 500 GB+ logs in OpenSearch, improving cross-team debug efficiency by 75%.
- Enforced AWS Control Tower guardrails across 12 accounts, resolving 300+ compliance violations in 3 months.
- Blocked 1,200+ malicious requests/day via AWS WAF rules and GuardDuty automation.
- Implemented GitHub Actions pipelines for 30+ repositories, saving 200+ engineering hours/month.

WORK EXPERIENCE

Senior Cloud & DevOps Engineer | Cloud Ripples Pvt Ltd, Chennai

Sep 2020 – Present

- Designed and operated multi-tier AWS network architecture: WAF → CloudFront (CDN/S3 frontend) → ALB → Nginx (ECS service) → AWS Cloud Map → backend microservices for secure, scalable request routing.
- Managed containerized workloads across Amazon ECS (Fargate & EC2) and Amazon EKS, deploying Java, Node.js, and Python microservices across dev, staging, and production environments.
- Authored and maintained Dockerfiles, ECS task definitions, and Kubernetes YAML manifests for consistent container lifecycle management.
- Built and maintained CI/CD pipelines with GitHub Actions, GitLab CI, Jenkins, and Argo CD for automated multi-project deployments using GitOps principles.
- Developed Terraform and CloudFormation IaC scripts for repeatable, versioned AWS resource provisioning.
- Productionized LLM inference: deployed Hugging Face models via vLLM (PagedAttention) for high-throughput serving; configured Ollama for rapid local GGUF model prototyping.
- Monitored cloud and containerized workloads using CloudWatch, Prometheus, Grafana, and EFK stack; configured Filebeat agents to stream container logs to OpenSearch.
- Implemented DevSecOps controls: GuardDuty automation, Security Hub, IAM least-privilege, Keycloak SSO, and AWS Control Tower multi-account governance.

DevOps Engineer | BambeeQ Software Solutions Pvt Ltd, Chennai

Mar 2020 – Aug 2020

- Automated database backups using Bash scripts with S3 integration; reduced storage costs via S3 lifecycle policies.
- Configured Apache and Nginx web servers with SSL/TLS on Linux; implemented CI/CD via Bamboo and AWS CodeDeploy.

Delivery Software Engineer | Mphasis Ltd (An HP Company), Bangalore

Apr 2010 – Aug 2014

- Handled incident management via HP Service Manager, Zenoss, and Business Availability Center within SLA targets.
- Maintained comprehensive documentation of environment strategies, processes, and support procedures.

PROJECT EXPERIENCE

Student Risk Management & Intervention Platform

2026– Present

- Provisioned ECS services behind ALB for containerized application hosting; integrated Keycloak for identity and access management with AWS Client VPN for secure private network access.
- Configured AWS Transfer Family (SFTP) for secure file ingestion workflows; managed PostgreSQL RDS for application data.
- Implemented observability using OpenObserve for log monitoring and Airbrake for error tracking; used GrowthBook for feature flagging and A/B testing.
- Automated infrastructure provisioning with Terraform and CI/CD deployments via GitHub Actions and ECR.

Technologies: AWS ECS, ALB, ECR, RDS (PostgreSQL), Client VPN, Transfer Family (SFTP), Keycloak, OpenObserve, Airbrake, GrowthBook, Terraform, GitHub Actions

AI-Powered Adaptive Learning Personalization Platform

2025– Present

- Architected multi-tier traffic flow: WAF → CloudFront (S3-hosted React SPA) → ALB → ECS → Nginx ECS service as target group → Auto Scaling → AWS Cloud Map → backend ECS microservices → PostgreSQL RDS.
- Deployed vLLM for production LLM inference (Hugging Face FP16 models) and Ollama for local GGUF model testing (Llama 3, Mistral), achieving 50% inference cost savings.
- Automated container image builds and deployments via GitHub Actions and ECR across all environments.

Technologies: AWS WAF, CloudFront, S3, ALB, ECS, Nginx, Auto Scaling, Cloud Map, ECR, RDS (PostgreSQL), vLLM, Ollama, GitHub Actions

Open-Source Competency-Based Adaptive Learning Platform

2021 – 2024

- Architected traffic flow: WAF → CloudFront (S3 frontend) → ALB (Load Balancer Controller) → Nginx Ingress Controller pod as target group → Auto Scaling → EKS microservices → PostgreSQL RDS.
- Managed full GitOps deployment pipeline using Jenkins, Argo CD, and ECR.
- Built complete observability stack: Filebeat → OpenSearch for centralised log management; Prometheus and Grafana for infrastructure and application metrics monitoring.

Technologies: AWS WAF, CloudFront, S3, ALB, EKS, Nginx Ingress, Auto Scaling, ECR, RDS (PostgreSQL), Jenkins, Argo CD, Filebeat, OpenSearch, Prometheus, Grafana

AWS Infrastructure for Biotechnology Application

- Designed VPC architecture with public/private subnets, route tables, NACLs, and security groups per environment; configured VPC peering via Transit Gateway.
- Established site-to-site VPN from two on-premises locations; set up AWS Client VPN endpoint for private resource access.
- Deployed FSx Lustre (persistent & scratch) across VPC using user data mount scripts, with S3 data repository for backup.
- Enabled full compliance stack: CloudWatch, CloudTrail, GuardDuty, Security Hub, AWS Backup, DataSync, Config, and VPC Flow Logs.

Technologies: AWS VPC, Transit Gateway, Site-to-Site VPN, Client VPN, FSx Lustre, CloudWatch, CloudTrail, GuardDuty, Security Hub, AWS Backup, DataSync, Config, VPC Flow Logs

Online Media Platform Application

- Provisioned EC2 instances for development with VPN-only access enforcement; managed IAM users and S3 bucket policies with fine-grained controls.
- Deployed CIS Benchmark v1.2.0 alarms via CloudFormation (AWS Foundations Benchmark); diagnosed Lambda performance issues; provided 24×7 infrastructure support.

Technologies: AWS EC2, Lambda, S3, IAM, Security Hub, CloudFormation, CloudWatch, GitLab

Healthcare Information System

- Managed Cognito user pools for authentication; analysed RDS connection pools and locked queries for performance tuning.
- Monitored API Gateway and Lambda logs for diagnostics; validated SNS delivery for email/mobile user verification; provided 24×7 support.

Technologies: API Gateway, Lambda, Cognito, SNS, RDS, S3, IAM, CloudFront, Route 53

Email Marketing Web Application

- Deployed and managed high-availability WordPress infrastructure; installed SSL/TLS (GoDaddy, Namecheap, Let's Encrypt); managed MySQL/PostgreSQL version upgrades and storage scaling.
- Configured auto-scaling with ALB; monitored via Nagios and CloudWatch; provided 24×7 support with root-cause analysis.

Technologies: Apache, Nginx, PHP, WordPress, MySQL, PostgreSQL, Route 53, ELB, EBS, CloudFront CDN, CloudFormation, Bash, Jira

School Management Application

- Set up Nginx/Apache web servers with SSL/TLS for PHP applications; configured CloudFront CDN with S3 origin; managed EC2 auto-scaling with ALB.

Technologies: Apache, Nginx, PHP, MySQL, Route 53, GitLab, CloudFront CDN

EDUCATION

Master of Computer Applications (MCA)

2001 – 2004

Velammal College of Management & Computer Studies, University of Madras

Bachelor of Science – Mathematics

1997 – 2000

Pachaiyappa's College, University of Madras